

**VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2008
Higher 2**

VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE

CANDIDATE NAME

CLASS

--	--	--	--	--

BIOLOGY

9747/01

Paper 1 Multiple Choice

19 September 2008

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.

Do not use any staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

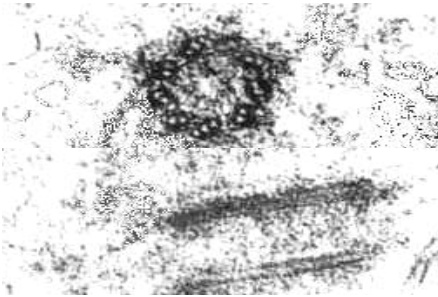
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet

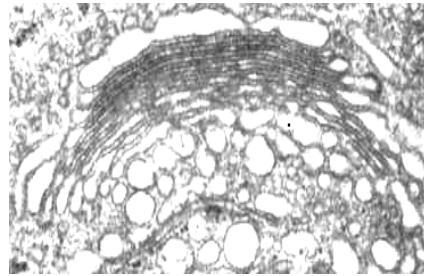
This paper consists of **19** printed pages.

- 1 The following contains images of a number of structures that can be seen in a cell.

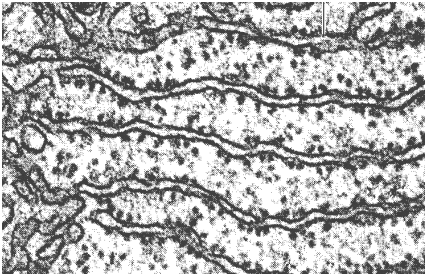
1



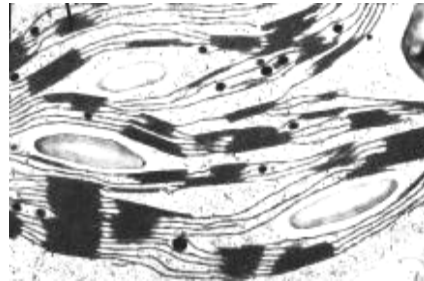
2



3



4

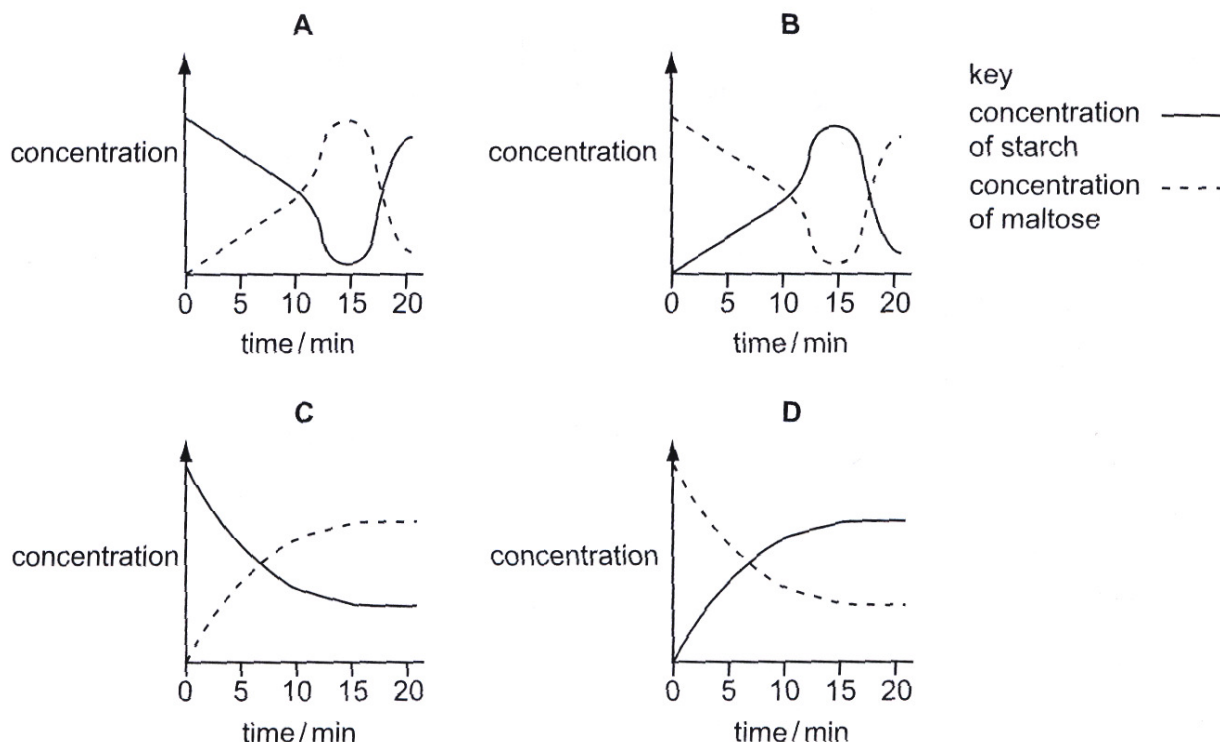


Identify structures 1 – 4.

	1	2	3	4
A	chloroplast	nucleus	ribosome	centriole
B	centriole	Golgi apparatus	rough endoplasmic reticulum	chloroplast
C	centriole	Golgi apparatus	rough endoplasmic reticulum	smooth endoplasmic reticulum
D	flagellum	Smooth endoplasmic reticulum	Golgi apparatus	chloroplast

- 2 Amylase is an enzyme that **breaks down amylose**. In an experiment using starch and amylase, the concentrations of starch and maltose present in the reacting mixture are measured every minute for 20 minutes. **After 10 minutes**, 1% hydrochloric acid is added and the mixture is heated to 60°C so that **acid hydrolysis** will occur.

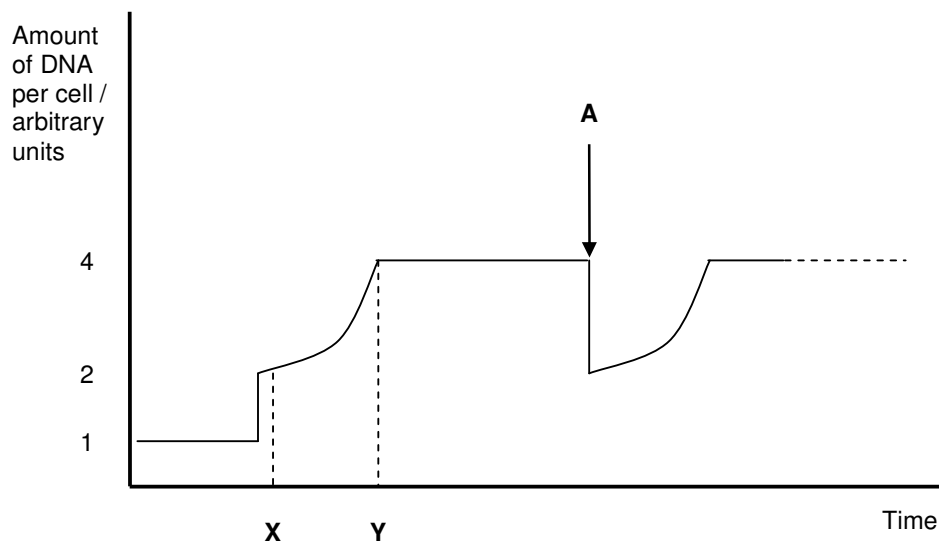
Which graph represents the results of this experiment?



- 3 Which of the following is/are the most likely consequence for a cell lacking functional Golgi bodies?
1. The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.
 2. The cell dies from a lack of enzymes to digest food taken in by endocytosis.
 3. The cell dies from the accumulation of worn-out organelles within itself.
 4. The cell is unable to reproduce itself
 5. The cell is unable to export its enzymes or peptide hormones.

- A** 1 and 5
B 2 and 3
C 1, 2, 3 and 4
D 1, 2, 3 and 5

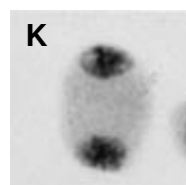
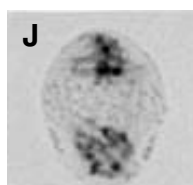
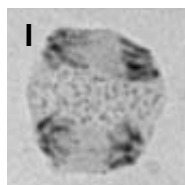
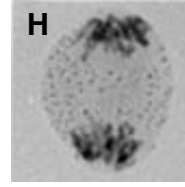
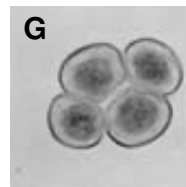
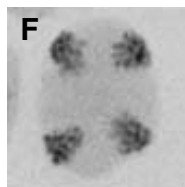
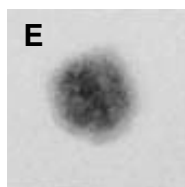
- 4 The graph below represents the changes in the quantity of DNA of a cell in the life cycle of an organism.



Which stage occurs between time **X** and **Y**, and at **A**?

	X and Y	A
A	Interphase	Cytokinesis
B	Prophase	Metaphase
C	Interphase	Anaphase
D	Prophase	Telophase

- 5 The micrographs (**E** to **L**) below show the appearance of cells from a plant anther. Arrange the order of the different stages of nuclear division.



- A** E → L → H → F → J → I → K → G
B E → L → H → K → J → I → F → G
C E → L → J → K → H → I → F → G
D E → J → I → F → L → H → K → G

- 6 How does the second meiotic division differ from mitosis?
In the second meiotic division
- A each chromosome replicates to form two chromatids during metaphase.
 - B exchange of genetic material occurs between chromatids.
 - C chiasmata form between the chromatids of a bivalent.
 - D the separating chromatids of a pair differ genetically.
- 7 Okazaki fragments are used to elongate
- A the leading strand toward the replication fork
 - B both strands in both directions
 - C the leading strand away from the replication fork
 - D the lagging strand away from the replication fork
- 8 Bacteria containing unlabelled DNA were transferred to a medium containing a radioactive label. What proportion of the double stranded molecules would have one labeled and one unlabelled strand after the third round of replication?
- A 1 of 2
 - B 1 of 4
 - C 1 of 8
 - D 1 of 16
- 9 Which of the following features is common to both DNA replication and transcription?
- A Nucleotides are added to the 5' end of the newly synthesized strand.
 - B Both RNA and DNA polymerase require oligonucleotide priming.
 - C Deoxyribonucleotides are incorporated into the growing sequence.
 - D Condensation reactions occur to form bond between the 3' hydroxyl and the 5' phosphate.
- 10 The RNA transcript, 5' AUGAUGAUGAUG 3', was generated using a synthetic DNA template. The letters **P**, **Q**, and **R** have been selected to represent different amino acids. Assume that each different group of codon codes for a different amino acid.
- If the genetic code required only two nucleotides for incorporation of amino acids, the amino acid sequence using this transcript would be
- A **P-R-Q-P-Q-R.**
 - B **P-P-P-P-P-P.**
 - C **P-Q-R-P-Q-R.**
 - D **P-R-P-R-P-R.**

11 Which characteristic of the genetic code lowers the likelihood of gene mutation?

- A** universality
- B** degeneracy
- C** triplet
- D** non-overlapping nature

12 How do viruses cause diseases in animals?

1. They inhibit normal host cell DNA, RNA, or protein synthesis.
2. They degrade the host cell's chromosomes.
3. They disrupt the oncogenes of the host cell causing uncontrolled cell division.
4. Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defenses.
5. They deplete the host cell of cellular materials essential for metabolic functions.

- A** 1, 2 and 5
- B** 1, 2, 3 and 5
- C** 1, 2, 4 and 5
- D** All the above

13 When a bacterial cell is infected by a lambda phage gets co-infected with another lambda phage at a later point in time, this may not result in the death of the bacterial cell. Which of the following statements is/ are possible mechanism/s for the phenomenon?

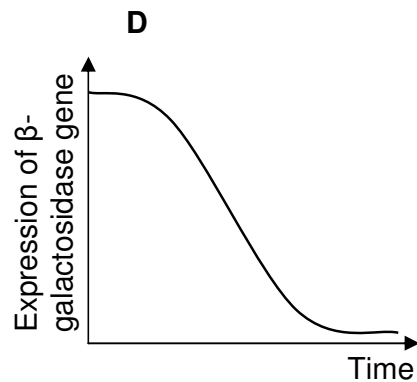
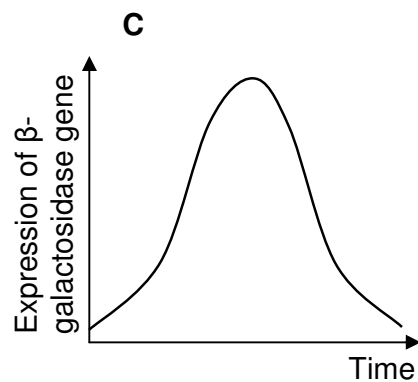
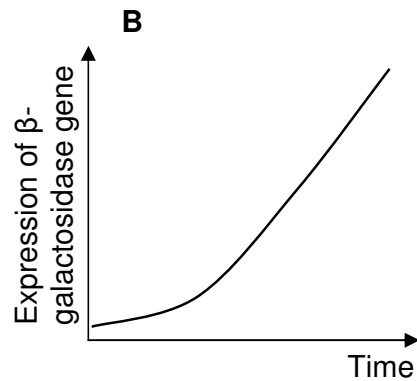
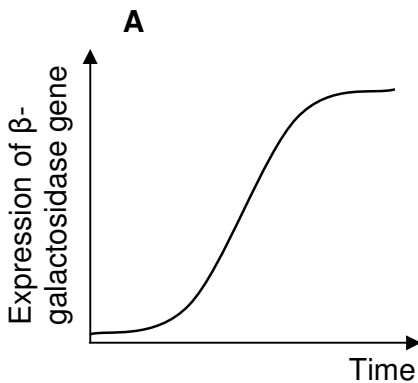
1. One of the viruses makes a larger number of new phages.
2. The transcription and translation of viral DNA is not initiated.
3. A repressor protein is synthesized to shut down replication of the phage.
4. Mutation of host genome results in the inability of lambda phage to enter host cell.

- A** 2 only
- B** 2 and 3
- C** 1 and 4
- D** 1 and 3

14 A bacterial cell that is able to take up naked DNA is said to be

- A liable
- B infected
- C competent
- D transformed

15 Both lactose and phenyl- β -D-galactose (phenyl-Gal) are substrates for β -galactosidase, but phenyl-Gal does not inactivate the lac-operon repressor and so is not an inducer. In an experiment, *E. coli* cells were first grown in lactose only and then in phenyl-Gal only. Which of the following graphs best represents the expression profile of β -galactosidase gene over the course of ten hours?

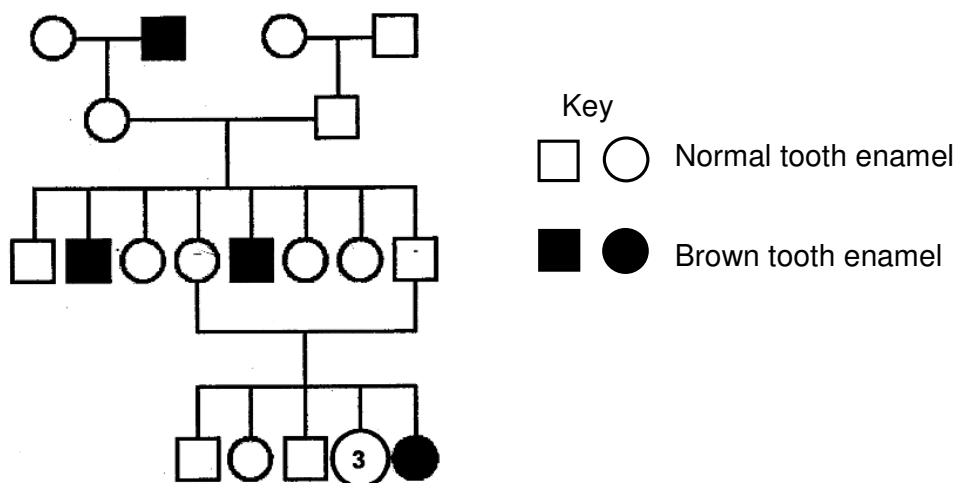


16 Enhancers are

- A proximal control elements where activators bind
- B distant control elements where regulatory proteins bind
- C proteins that bind with repressors, deactivating them
- D sites where RNA polymerase bind

- 17 Which of the following is a feature of prokaryotic gene expression?
- A Monocistronic mRNA, rather than polycistronic mRNA, is produced.
 - B RNA synthesis and protein synthesis are coupled.
 - C Many genes are interrupted by non-coding sequences.
 - D mRNA is often extensively modified before translation
- 18 Which of the following provides a possible mechanism for the production of a wide number of types of antibody proteins from a small number of genes?
- A gene amplification
 - B alternative splicing
 - C crossing over and random segregation
 - D deletions and random transposition of DNA segments
- 19 All somatic nuclei of an eukaryotes contain
- A only euchromatin except in the case of the Y-chromosome.
 - B genes to specify the portion of the organism in which they are found.
 - C all the information needed for growing the whole organism.
 - D all of the chromosomes except sex chromosomes which are restricted to sex organs.
- 20 If a proto-oncogene mutated into an oncogene in an otherwise normal cell, the immediate result would likely be
- A apoptosis.
 - B arrest of the cell cycle.
 - C a slight quickening of the cell cycle.
 - D rapid mutations of other genes.
- 21 Two unlinked loci affect mouse fur colour. **CC** or **Cc** mice are agouti. Mice with genotype **cc** are albino because all pigment production and deposition of pigment in hair is blocked. At the second locus, the **B** allele (black agouti coat) is dominant to the **b** allele (brown agouti coat).
- A mouse with a black agouti coat is mated with an albino mouse of genotype **bbcc**. Half of the offspring are albino, one quarter are black agouti, and one quarter are brown agouti.
- What is the genotype of the black agouti parent?
- A **BB CC**
 - B **Bb CC**
 - C **BB Cc**
 - D **Bb Cc**

- 22 The diagram below shows the inheritance pattern of tooth enamel in chimpanzee.



If individual 3 mates with a partner with the brown tooth enamel, what is the probability that their first offspring will be a female but without the brown tooth enamel?

- A $1/2$
- B $1/3$
- C $1/4$
- D $1/6$
- 23 In the magpie moth *Abbraxas* sp., the female is the heterogametic sex and the gene for wing colour is sex-linked. From a cross between a normal coloured male and a pale coloured female, the F₁ offspring consisted of all normal coloured individuals with the two sexes in equal proportions. Which ratio would be obtained in the F₂ generation produced from these?
- A Normal coloured males to normal females 1:1
- B Normal coloured males and females to pale females 3:1
- C Normal coloured males and females to pale males and females 1:1
- D Normal coloured males to pale coloured females 1:1

24 Which of the following statements are true?

- 1 Continuous variation is controlled by the additive effects of polygenes.
- 2 Continuous variation is always affected by the environment.
- 3 Discontinuous variation is sometimes affected by the environment.
- 4 Discontinuous variation exhibits a normal distribution curve.

- A 1 and 2
 B 1 and 3
 C 3 and 4
 D None of the above

25 The light reaction of photosynthesis does not include

- A chemiosmosis
 B pigment photoactivation
 C NAD reduction
 D electron transport

26 The blue dye DCPIP can be converted to colourless DCPIP as shown below:

DCPIP (blue) → reduced DCPIP (colourless)

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following ways.

Tube	Contents	Treatment	Colour	
			At start	After 20 minutes
1	1 cm ³ chloroplast suspension + 5 cm ³ DCPIP	Illuminated strongly	Blue green	Green
2	1 cm ³ buffer solution + 5 cm ³ DCPIP	Illuminated strongly	Blue	Blue
3	1 cm ³ chloroplast suspension + 5 cm ³ DCPIP	Left in the dark	Blue green	Blue green

Which one of the following statements is a possible conclusion for the observation above?

- A Photolysis of water produces oxygen which oxidizes DCPIP.
 B Respiration consumes oxygen from the suspension, hence DCPIP is partially reduced.
 C Light reaction which occurs in the chloroplasts yield free electrons which reduced DCPIP.
 D Either strong illumination or the buffer solution used in the extraction of chloroplasts could oxidize DCPIP.

- 27 A field of wheat had become infected by the yellow rust fungus, *Puccinia striiformis*, which attacks the leaves. To investigate the effect of infection on yield, 10 ears of wheat were collected when mature from healthy plants and plants with leaves infected. The dry mass (in grams) of the grain from each ear was determined and is given below. The mean ($\bar{x}$) and standard deviation (s) are given for each sample.

Dry mass of grain in each ear/ g	
Healthy plants	Infected plants
4.01	3.76
4.15	3.09
4.13	3.27
3.96	3.56
4.20	3.44
4.17	3.18
4.03	3.60
4.12	3.39
3.88	3.27
4.03	3.45
$\bar{x} = 4.07$	$\bar{x} = 3.40$

A t-test was used to find out about the effect of infection by *Puccinia striiformis* on yield. The formula is given below:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{x}$ = mean
 s = standard deviation
 n = sample size
 x = observation
 Σ = 'sum of'

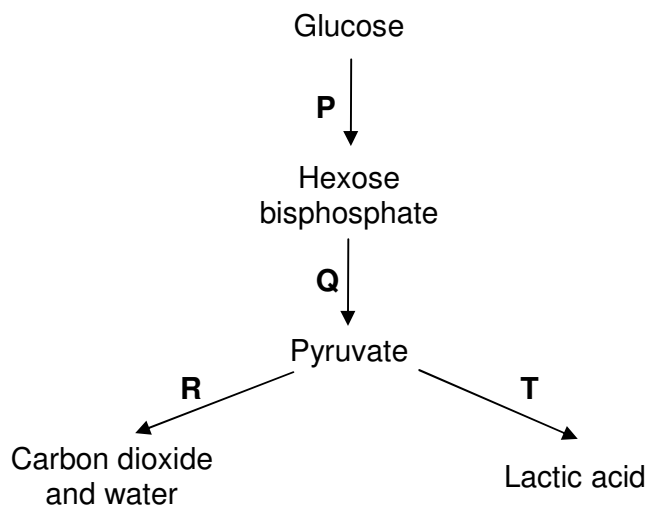
$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Degrees of freedom	t-values at corresponding p-values			
	p = 0.1	p = 0.05	p = 0.01	p = 0.001
8	1.86	2.31	3.36	5.04
9	1.83	2.26	3.25	4.78
10	1.81	2.23	3.17	4.59
11	1.80	2.20	3.11	4.44
12	1.78	2.18	3.05	4.32
13	1.77	2.16	3.01	4.22
14	1.76	2.15	2.98	4.14
15	1.75	2.13	2.95	4.07
16	1.75	2.12	2.92	4.02
17	1.74	2.11	2.90	3.97
18	1.73	2.10	2.88	3.92
19	1.73	2.09	2.86	3.88
20	1.72	2.09	2.85	3.85

Which of the following conclusions made about the effect of infection by *Puccinia striiformis* on yield is **correct**?

- A** Since $p < 0.05$, the difference between the dry mass of grain from each ear taken from the healthy plants and infected plants is significant. Thus infection by *Puccinia striiformis* has no effect on yield.
- B** Since $p < 0.05$, the difference between the dry mass of grain from each ear taken from the healthy plants and infected plants is not significant. Thus infection by *Puccinia striiformis* has an effect on yield.
- C** Since $p < 0.001$, the difference between the dry mass of grain from each ear taken from the healthy plants and infected plants is significant. Thus infection by *Puccinia striiformis* has an effect on yield.
- D** Since $p > 0.001$, the difference between the dry mass of grain from each ear taken from the healthy plants and infected plants is not significant. Thus infection by *Puccinia striiformis* has no effect on yield.

28 To which of the arrows in the figure below do the following statements apply?



- I.** NAD^+ is regenerated without the use of the electron transport system.
- II.** ATP is synthesized via substrate level phosphorylation.
- III.** It can take place under aerobic conditions.

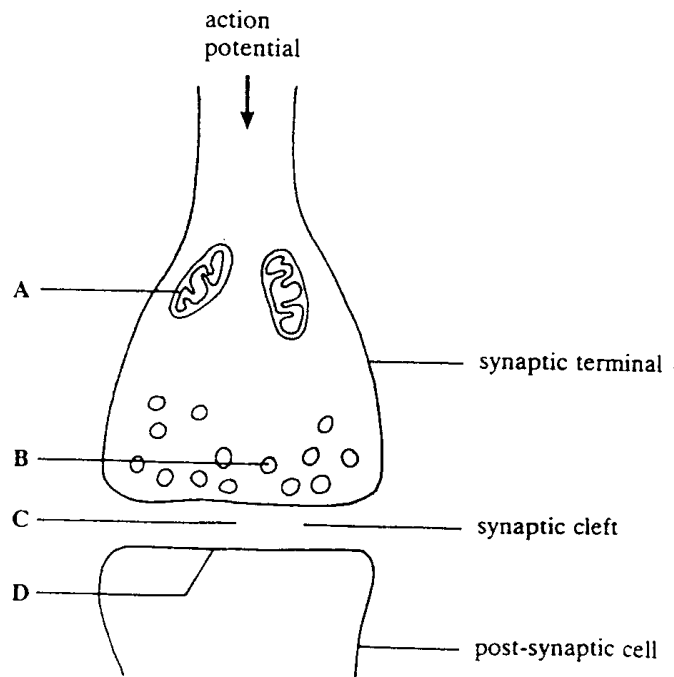
	I	II	III
A	Q	R	T
B	P	R	T
C	T	Q	P
D	T	Q	R

- 29** Anesthetics reduce pain by blocking the transmission of nerve impulses. Which of the following might work as an anesthetic?
- A** A chemical that blocks voltage-gated sodium channels in membranes.
 - B** A chemical that opens voltage-gated potassium ions.
 - C** A chemical that blocks neurotransmitter receptors.
 - D** All of the above.

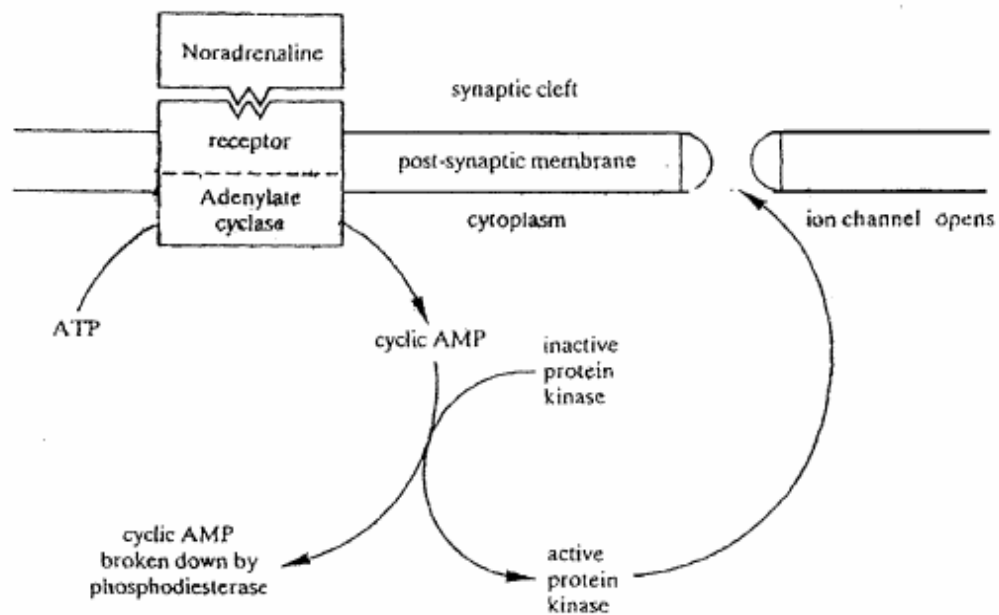
Poisonous snakes produce a toxin which affects synaptic transmission. It does not affect acetylcholine production and release, or the enzyme acetylcholinesterase but it does produce irreversible paralysis and death.

At which site, **A – D**, in the diagram does the toxin act?

Do you want this question instead?



- 30** Noradrenaline stimulates the activity of an enzyme adenylate cyclase located on the post-synaptic membrane of a neurone. This initiates the sequence of reactions shown in the diagram, causing the opening of channels in the membrane through which ions can pass.



Caffeine causes the ion channels to remain open. A possible explanation is the inhibition of

- A** ATP production
- B** phosphodiesterase
- C** cyclic AMP production
- D** adenylate cyclase

31 Which of the following statements describe genetic drift?

1. Genetic drift does not require the presence of variation.
2. Genetic drift involves competition between members of a species.
3. Genetic drift maintains the allele frequencies within a gene pool.
4. Founders effect is an example of genetic drift.

- A** 1 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 3 and 4 only

32 Which of the following is an example of a clade?

- A** The first fish species and every living organism that looks like a fish.
- B** The first dicotyledonous plant species and all its descendants.
- C** The first bird species and its most recent descendants only.

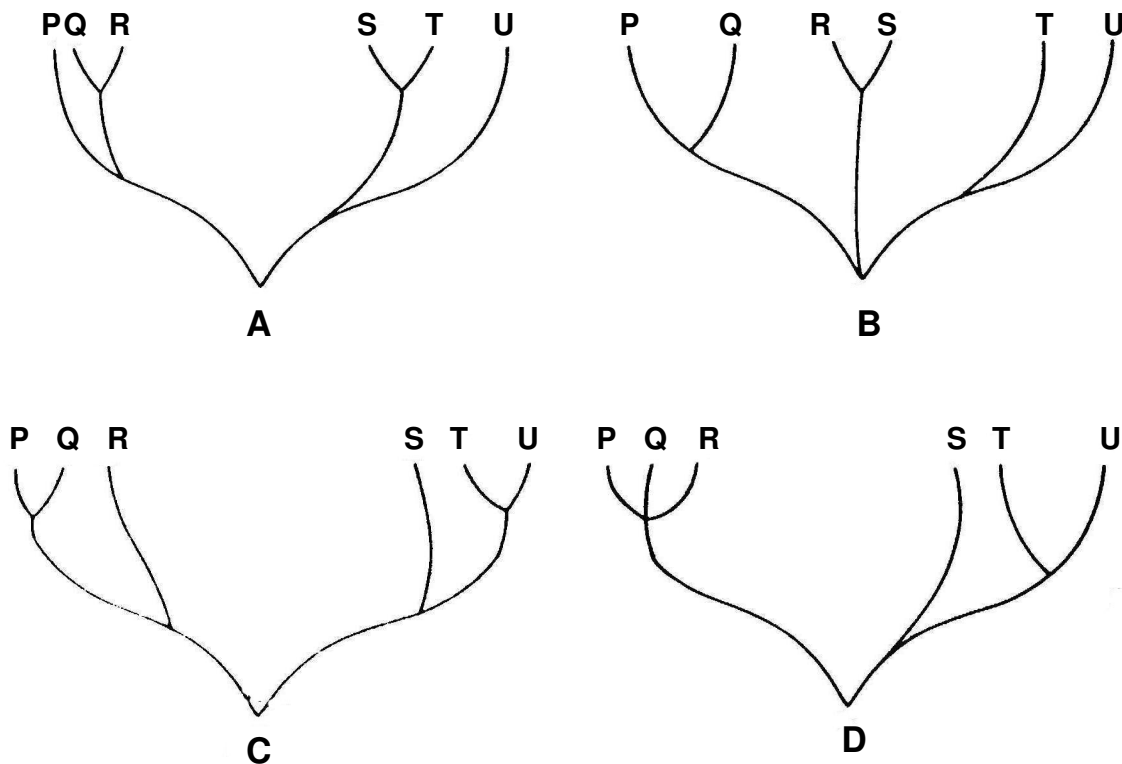
D All the descendents of reptiles excluding their common ancestors.

- 33** Cytochrome c oxidase is an enzyme found in all aerobic organisms as part of their respiratory metabolism. As a result of mutation and reproductive isolation, the amino acid composition of the enzyme varies between species.

The number of amino acid **differences** between species P, Q, R, S, T and U are shown below:

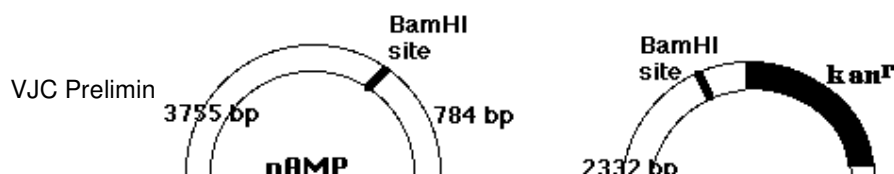
	P	Q	R	S	T	U
P	0	4	11	21	23	22
Q	4	0	9	19	22	24
R	11	9	0	15	17	20
S	21	19	15	0	10	9
T	23	22	17	10	0	2
U	22	24	20	9	2	0

Which one of the following diagrams is consistent with the evolutionary relationships between the species suggested by these amino acid differences?



Refer to the diagram below for Questions **34** and **35**.

Two types of plasmids, pAMP and pKAN are illustrated below. pAMP carries an ampicillin-resistant gene (amp^r) and pKAN carries a kanamycin-resistant gene (kan^r).



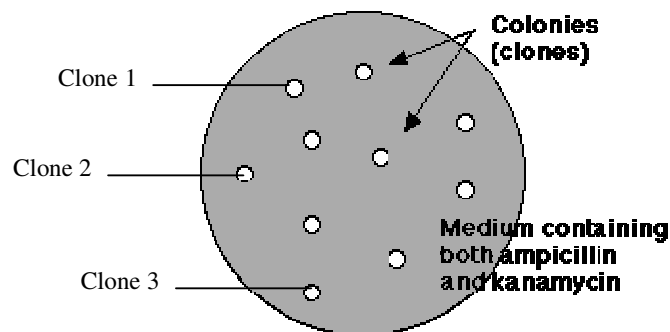
- 34** Solutions of these two plasmids are mixed together with the enzymes BamHI and HindIII. The pAMP and pKAN fragments were then allowed to ligate in the presence of DNA ligase.

Which of the following represent rejoined molecules could be used as a vector in DNA cloning?

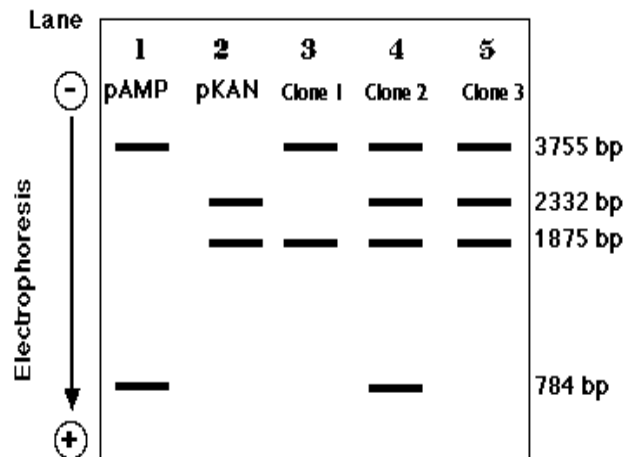
1. 1875 bp
2. 2659 bp
3. 3116 bp
4. 5630 bp

- A** 4 only
B 1 and 3
C 3 and 4
D 2, 3 and 4

- 35** Treatment of *E. coli* with the mixture of resultant molecules formed in Question **34** produces some colonies that are able to grow in the presence of both ampicillin and kanamycin.



Plasmid from clones 1, 2 and 3 were isolated and treated with enzymes BamHI and HindIII. Electrophoresis was carried out on these samples with results as shown in the diagram below.



Which of the following statements are true?

1. Clone 1 is transformed by a recombinant plasmid
2. Clone 2 is transformed by both re-annealed pKAN plasmid and re-annealed pAMP plasmids.
3. Clone 3 is transformed by the recombinant plasmid as well as the re-annealed pKan plasmid.

- A** 1 only
B 2 only
C 1 and 2
D 1, 2 and 3

36 Which of the following statement is true?

Genomic library

- A** Bacteria plasmid used as vector.
B Fragments contain introns.
C Used for studying genes responsible for the specialised functions of a particular cell type.
D Represents a sample of genes expressed at a certain time in the life of an organism.

cDNA library

- Bacteriophage used as vector.
 No introns.
 Used for studying alleles of a particular gene.
 Represents a random sample of all the DNA sequences in an organism.

- 37** The restriction enzyme *Ddel* recognizes a sequence on DNA, CTNAG (where N is any base). A portion of the β -hemoglobin sequence encoding normal hemoglobin is CCTGAGGAG. If the mutation leading to sickle cell anemia had been CCTCAGGAG, would you be able to distinguish the two alleles with a restriction digest?
- A** Yes, the normal allele will be digested, the sickle allele will not.
 - B** Yes, the sickle allele will be digested, the normal allele will not
 - C** No, both will be digested by *Ddel*.
 - D** No, neither will be digested by *Ddel*.
- 38** Hematopoietic stem cells used in gene therapy can be obtained from
- A** the embryo.
 - B** the umbilical cord blood.
 - C** the bone marrow.
 - D** the umbilical cord blood and bone marrow.
- 39** Duchenne muscular dystrophy (DMD) is a common lethal X-linked human genetic disease caused by the absence of the protein dystrophin in muscle fibres. Gene therapy using recombinant adenoviruses with part of the normal allele of the gene coding for dystrophin into a muscle did not provide a permanent cure for DMD. Trials showed that the transgene expression declined after about 2 weeks and was negligible after only 4 weeks.
- Which of the following statements is/are possible explanation(s) of this phenomenon?
- 1. Rejection of the vector by the host immune system.
 - 2. Failure to integrate target gene into the genome of muscle cells.
 - 3. Post-transcriptional modification could be missing.
 - 4. Unequal segregation of transgene into myofibrils when cells divide.
 - 5. Replacement of worn-out muscle cells with non-functional cells.
- A** 1, 2 and 3
 - B** 1, 3 and 4
 - C** 2, 4 and 5
 - D** 3, 4 and 5

40 Which of the following is a mismatch between the process and its outcome?

	Process	Outcome
A	Anther culture	Production of heterozygous plants
B	Embryo culture	Production of distantly related plants
C	Protoplast culture	Formation of genetically identical plants
D	Callus culture	Generation of differentiated tissues